Causal Peer effect Analysis for Irreversible Decisions

Vincent Starck*†
LMU Munich

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Abstract

I study peer effects that arise from irreversible decisions in the absence of a standard social equilibrium. I model a latent sequence of decisions in continuous time and obtain a closed-form expression for the likelihood, which allows estimation of the proposed causal estimands under a version of unconfoundedness and overlap. The method avoids linear-in-means regression by modeling the (possibly unobserved) realized direction of causality, whose probability is identified. I provide identification and inference results with either several networks or one large network, while allowing for heterogeneity in peer effects.

Keywords: Peer effects, Continuous time, Heterogeneity, Causal Inference, Networks.

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†V.Starck@lmu.de

1 Introduction

Analyzing peer effects is notoriously difficult. In a seminal paper, Manski (1993) discusses the reflection problem that arises when one runs a regression on the conditional expectation, which induces restrictions on the regression coefficients that lead to tautological models or identification issues.

In applications that feature small group or friendship networks, the channel of influence plausibly operates through actual outcomes, which leads to Spatial Autoregressive (SAR) or linear-in-means (LIM) models in the outcomes. They alleviate reflection problems insofar as they allow for identification and estimation of peer effects (Bramoullé, Djebbari and Fortin, 2009; De Giorgi, Pellizzari and Redaelli, 2010; Blume et al., 2015; Martellosio, 2022), *e.g.*, through the use of non-overlapping peer groups that instrument away the simultaneity bias.

However, inference about the effect of peers using SAR models can be challenging. Manski (1993) points out the difficulties in identifying and estimating peer effects that transcend reflection regression issues. Identification can be tenuous in SAR models as it is based on functional form restrictions that are particularly sensitive to, *e.g.*, measurement error or mechanical relationships between individual outcomes and group means (Gibbons and Overman, 2012; Angrist, 2014). Although identification is typically the rule when the network is known (Blume et al., 2015), identification is dependent on the mechanics of the model through its reduced form. Moreover, reliable inference about peer effects comes with caveats even in the absence of identification problems (Hayes and Levin, 2024; Wang and Jadbabaie, 2025) and a causal interpretation of the contagion coefficient may be lacking (Debarsy and Le Gallo, 2025).

Focusing on the case of irreversible decisions, I develop a new framework by modeling the latent sequence of decisions in continuous time. In this case, the standard notion of social equilibrium[1] is inadequate as people opt in over time according to their characteristics and shocks, then cannot subsequently adjust their outcomes. This setup offers a natural framework to discuss causality, as it breaks the simultaneity by separating first-movers that potentially generate a causal reaction from the subjects of influence. I formalize the counterfactual framework using potential

[1]The solution for the conditional expectation or the vector of outcomes implied by the linear-in-means specification.

outcomes and propose a counterpart to the usual unconfoundedness and overlap assumptions for identification. Then, I show that causal peer effect parameters lend themselves to straightforward inference via maximum likelihood.

In this context, key information lies in the order in which individuals select into the absorbing state. When unknown, it is potentially an object of interest. Although the realized order cannot be identified, the probability of any order can be identified as a function of individuals' network and characteristics; I provide a framework to estimate those probabilities. Conversely, when the order is (partially) known, it carries identifying power that is useful for identifying and estimating peer effects without restricting heterogeneity, functional form, or the presence of contextual and correlated effects.

As we distinguish between sources and recipients of peer influence, exploring peer-effect heterogeneity can be more fruitful, as it provides parameters that are more natural to interpret. Heterogeneity in peer effects is likely relevant, but remains underexplored (recent work on the subject includes Masten (2018); Mogstad, Torgovitsky and Volpe (2024)). A source of heterogeneity may be that the influence of i on j need not be the same as that of j on i, whence the relevance of orderings, or that people may react to different friends in different ways. For instance, people may react differently upon observing a popular or highly educated peer opting in than this peer would in the converse situation. Some of the characteristics that describe heterogeneity of responses also make people move first with a higher probability.

I consider two main regimes: one large network and many networks. Large networks with observed ordering are found to be fully identified, *i.e.*, peer effects of heterogeneous forms are identified separately from influence of covariates under the maintained exogeneity conditions. Identification stems from information about the order of moves and is not tied to functional form restrictions; no aspect of heterogeneity needs to be restricted.

In the many, small network case, identification may be more tenuous because of incidental parameters if correlated effects are present, which are especially problematic if those effects can interact with any other covariate or peer effect strength. In this case, they must be handled by a combination of functional form restrictions, distributional assumptions, and extra information. On the other hand, in the absence of correlated effects or additional restrictions, identifying heterogeneous, nonlinear peer

effects and the probability of orderings is feasible.

The paper contributes to the literature on peer effects, in particular identification and estimation issues (Manski, 1993; Bramoullé, Djebbari and Fortin, 2009; Angrist, 2014; Blume et al., 2015; Hayes and Levin, 2024). Frameworks for analyzing peer effects beyond linear models are scarce but often desirable. As summarized in Sacerdote (2014), "researchers have shown that linear-in-means model of peer effects is often not a good description of the world, although we do not yet have an agreed-upon model to replace it." Boucher et al. (2024) provide a recent breakthrough that extends the response class to contain mean, maximum, and minimum outcome in situations of equilibrium.

The paper is also connected to the literature on (interdependent) durations (Abbring and Van den Berg, 2003; De Paula, 2009; Honoré and De Paula, 2010; Honoré and de Paula, 2018; Lin and Liu, 2021). Abbring and Van den Berg (2003) establish identification of treatment effects in duration models using information on event timing. De Paula (2009); Honoré and De Paula (2010); Lin and Liu (2021) study stopping games in which individuals' exit times depend on the decisions of others. By contrast, I focus on the causal inference aspect of peer effects with limited structure. Although additional assumptions on people's behavior can provide insight and can be helpful when features such as synchronization are important (De Paula, 2009), the present approach focuses on the minimal assumptions for causal inference.

In addition, the results on orderings make connections to the literature on targeting and diffusion (Banerjee et al., 2013; He and Song, 2018) while the method may also be useful to model staggered treatment adoption (Shaikh and Toulis, 2021; Athey and Imbens, 2022) by relaxing the common assumption that treatment adoptions arise independently.

2 Causality, stochastic process, and likelihood

2.1 Setup and causal peer effect parameters

I focus on irreversible decisions: there is an initial default state, labeled 0, and the decision to opt-in leads to an absorbing state, labeled 1. The decision can be irreversible because it is physically impossible to undo (*e.g.,* vaccination), because it is costly to reverse in the timeframe considered (*e.g.,* migration decision), or because the focus is

on first-time events. The goal is to model and estimate peer effects, *i.e.,* how decisions of peers alter an individual's probability of opting in. The peers are described by a network; their importance is given by a weighting matrix W. Although it is naturally interpreted in what follows as a row-normalized matrix that gives positive weights to the peers of each individual, the results apply to general matrices provided that W is bounded in row and column sums.

The adoption time of an individual i, denoted by T_i, is a random variable that depends on individuals' characteristics, expectations, and peer effects. Observing a peer opting in may change the likelihood that an individual does. This can be due to conformity, information transmission, or other types of social influence. This may imply a change in the distribution of the adoption time of individual i, T_i, upon observing the adoption of a peer.

Adapting the potential outcome notation (Neyman, 1923; Rubin, 1974) to the current setup, the adoption time is represented by $T_i(\tau)$ for $\tau \in \mathbb{R}^{N-1}$: the adoption time of individual i may depend on the adoption times of other individuals. This dependence carries over to the outcomes $\mathbb{1}_{T_i \leq t}$, which are observed for $t = S > 0$: $y_i \stackrel{\text{def}}{=} \mathbb{1}_{T_i \leq S}$.

The changes in the distribution of outcomes following a change in τ can be interpreted as a peer effect. The causal impact can be captured by parameters that summarize the effect of a change in τ on the distribution. A parameter of interest is the analog of the average treatment effect: the mean change in the outcome following a change in τ:

$$\delta(\tilde{\tau}, \tau) \stackrel{\text{def}}{=} \mathbb{E}[Y_i(\tilde{\tau}) - Y_i(\tau)] = \mathbb{P}[Y_i(\tilde{\tau}) = 1] - \mathbb{P}[Y_i(\tau) = 1] \tag{1}$$

For instance, one could be interested in comparing $\tilde{\tau} = \vec{0} \in \mathbb{R}^{N-1}$ to $\tau = \vec{\infty}$ so that the parameter describes the change in adoption probability induced by the immediate adoption of all peers compared to them never adopting.

We could also be interested in counterfactual effects such as the expected adoption time, $\mathbb{E}[T_i]$, and various conditional versions, or the expected time before a fraction of the population opts in.

2.2 Stochastic process and unconfoundedness

Individual adoption is modeled with the following continuous-time stochastic process:

Definition 1 (Stochastic process)**.** Let T_i^1 be a set of random variables over $\mathbb{R}^+$. When a first individual, say j, opts in (*i.e.*, $j = \arg\min_i(T_i^1)$), she withdraws and the remaining random variables are updated to T_i^2, whose distribution may depend on the time elapsed and the event that j opted in. The process then goes on with the updated distributions, and so on. The time to adoption is therefore $T_i = \sum_{k=1}^{K(i)} T_{p_k}^k$, where the sum ranges from 1 to the round $K(i)$ where i comes first and p_k is the stage k 'winner'. The outcomes are observed at time S: $y_i = \mathbb{1}_{T_i \leq S}$.

The process generates a collection $T_i^* = (T_i^1, \ldots, T_i^N)$ of *latent partial times*, which are (conditionally on the result of each step) independent. This process is quite general in terms of the dynamics it allows. It basically assumes the arrow of time, ruling out feedback from the future.

Peer effects occur when the event that a friend opts in modifies the time distribution. Latent partial times thus incorporate distributional changes due to peer effects, but can also be correlated because characteristics influence both the choice of peers and the distribution of times, inducing, *e.g.*, homophily bias (Shalizi and Thomas, 2011). Controlling for these characteristics can restore independence:

Assumption 2 (Independence of latent partial times)**.** The latent partial times are conditionally independent: $\forall i, k,\ T_{-i}^{k+1} \perp\!\!\!\perp T_i^{k+1} | X, W, \mathcal{F}_k$,

where T_{-i} is the set of times of all individuals but i and $\mathcal{F}_k$ is the relevant filtration (collecting the previous $\arg\min_i T_i^k$ and $\min_i T_i^k$, *i.e.*, the identity of previous movers and their (partial) times).

The independence of latent partial times is related to a notion of unconfoundedness. Note that the potential outcomes can be expressed in terms of the latent partial times T_i^* *via*

$$T_i(\tau) = \sum_k \left(\mathbb{1}(T_i^k < \tau_{p_k} - \tau_{p_{k-1}}) T_i^k + \mathbb{1}(T_i^k \geq \tau_{p_k} - \tau_{p_{k-1}})\ (\tau_{p_k} - \tau_{p_{k-1}}) \right) \qquad (2)$$

where $(\tau_{p_1}, \ldots, \tau_{p_{N-1}})$ is the ordered version of τ, $\tau_{p_0} \stackrel{\text{def}}{=} 0$, $\tau_{p_N} \stackrel{\text{def}}{=} \infty$, the sum ranges from 1 to $\arg\min_k \{T_i^k < \tau_{p_k} - \tau_{p_{k-1}}\}$, and the latent partial times are based on τ (imposing that counterfactual histories agreeing with the realized history through

stage k generate the same latent partial times up to stage $k+1$). The observed time T_i then corresponds to $T_i(\tau)$ with $\tau_{p_k} - \tau_{p_{k-1}} = \min_{j \neq i} T_j^k$.

Then, unconfoundedness can be stated as follows.

Assumption 3 (Unconfoundedness of timings)**.** Potential outcomes are independent of others' latent times: $\forall i, k, \tau,\ T_{-i}^{k+1} \perp\!\!\!\perp T_i(\tau) | X, W, \mathcal{F}_k$, where the potential choices of τ are consistent with the information set $\mathcal{F}_k$.

Then, the following result follows from the formulation of potential outcomes (*e.g.*, setting $\tau_j = \infty$ for the people who have not opted in according to $\mathcal{F}_k$):

Proposition 1. *Unconfoundedness of timings implies independence of latent partial times.*

Unconfoundedness ensures that some (changes in) parameters have a causal interpretation. In the absence of peer effects, the potential outcomes do not vary with τ. Unconfoundedness states that network links are not predictive of unobserved factors that influence the time when people opt in. This requires controlling for variables that affect both network formation and the probability of participation, which is a strong assumption, but similar to frequently invoked exogeneity assumptions, *e.g.*, Bramoullé, Djebbari and Fortin (2009).

Although beyond the scope of this paper, modeling network formation to identify or control for latent variables (Goldsmith-Pinkham and Imbens, 2013; Graham, 2017; Auerbach, 2022; Starck, 2025) may help in the event of unobserved confounders. When feasible, randomization of peer groups provides a natural way to ensure unconfoundedness.

Finally, causal inference usually requires a sort of overlap assumption. In this context, it means that there is a non-trivial probability of observing the distributions in different states at each point in time. A simple way to ensure this is to assume that the distributions are fully supported over the positive real line.

Assumption 4 (Overlap)**.** The latent partial times have positive densities over $\mathbb{R}^+$.

This assumption can be relaxed in various ways, at the expense of more structure on individual behavior or peer influence. For instance, this formulation of overlap implies that simultaneity almost surely does not occur. With some restrictions on

the combined influence of peers to handle the effect on the remaining individuals, one could allow for a form of asymmetric simultaneity: when i opts in, j inherits a degenerate distribution to induce $T_i = T_j$.

In general, people simultaneously opting in may be an important component of some application (De Paula, 2009). Its relevance can be empirically assessed when some information about first times is available, as in the empirical application of the last section.

Remark 1 (Beyond binary outcomes)**.** The process could be extended to account for more complex decisions. For instance, there could be a set of absorbing states to select into or there could be a gradation in behavior (say, $y_i \in \mathbb{N}$ is non-decreasing over time). The analysis can be extended to such frameworks at the expense of additional notation and suitable regularity conditions.

2.3 Likelihood

Absent information about the times when people opt in, the latent distributions are not identified. To make the model tractable and easy to interpret, I specify the distribution to be exponential and assume that the rates depend on the peers who previously opted in:

Assumption 5. $T_i^{k+1}|X, W, \mathcal{F}_k \sim \text{Exp}(\lambda_i(X, W, \mathcal{F}_k))$.

I focus on Exponential distributions for two reasons. First, exponential waiting times arise automatically under the assumption of a constant probability per unit of time, a natural point of departure. Second, exponential distributions are particularly attractive from an analytical standpoint and ensure tractability. Due to the memorylessness property, conditioning on elapsed time is irrelevant. If we assume that the peer effects depend on the identity of the previous movers (regardless of their order), the relevant filtration consists only of a set with the identity of the previous movers. In the absence of peer effects, the rates simply do not evolve: $\lambda_i^{+\mathcal{F}_k} = \lambda_i$ for all i, k. In what follows, I let $\lambda_i^{+\mathcal{F}_k} \equiv \lambda_i(X, W, \mathcal{F}_k)$ with $\lambda_i \equiv \lambda_i^{+\emptyset}$.

Importantly, exponential rates can be left unrestricted as functions of covariates. Given a distribution of the outcomes, they are nonparametrically identified under few conditions. As such, the exponential specification provides a convenient framework in

which peer effects are easy to interpret, while allowing for considerable heterogeneity. Under the information structure considered here (outcomes at time S, possibly order of adoptions), the exponential specification provides a fit to the latent time dynamics that facilitates interpretation without letting peer effect identification depend on functional form. The assumption has more bite when time dynamics is explicitly used for identification or when time analysis is of interest (*e.g.*, diffusion analysis or generation of counterfactuals), because the exponential distribution over time then influences the estimates.

Remark 2. It is possible to let the rates depend on other pieces of information that are communicated at adoption times. For instance, if people disclose some information about their happiness or success with the outcome (*e.g.,* a colleague felt happy vs. bored after retiring or a friend did vs. did not have strong side effects from a vaccine), this can be included as a factor that affect the rates. This is a promising avenue to understand the mechanisms underlying peer effects, though it may constitute a strong data requirement.

The exponential rates capture the heterogeneity between individuals, and their changes over time reflect the influence of peers. Their changes can be directly interpreted as peer effects insofar as they capture the change in the shape of the distribution and the measure average reduction in adoption times. They also translate into practical formulae to compute parameters of interest.

Example 1. Consider the change in probability of adoption:

$$\mathbb{E}[Y_i(\vec{0}) - Y_i(\vec{\infty})|X] = e^{-\lambda_i S}(1 - e^{-(\lambda_i^+ - \lambda_i)S}), \tag{3}$$

where + corresponds to the final level of peer effects – all peers have previously opted in.

To first order, this simplifies to

$$e^{-\lambda_i S}(\lambda_i^+ - \lambda_i)S \tag{4}$$

The first term, $e^{-\lambda_i S}$, is the baseline probability of not opting in before time S, while $(\lambda_i^+ - \lambda_i)$ reflects the change in exponential rates induced by the prior adoption of friends.

The following examples consider simple homogeneous cases to illustrate the framework and build intuition for identification conditions, which are formalized later.

Example 2. Consider two (connected) individuals ($i = 1, 2$) and homogeneous rates λ, λ^+. The probabilities for the four possible outcomes are derived in the appendix. They are given by

$$\begin{aligned}
p_{00} &\stackrel{\text{def}}{=} \mathbb{P}[y_1 = y_2 = 0 | W_{12} = 1] = e^{-2\lambda S} \\
p_{10} &\stackrel{\text{def}}{=} \mathbb{P}[y_1 = 1, y_2 = 0 | W_{12} = 1] = \lambda e^{-\lambda^+ S} g(2\lambda - \lambda^+) \\
p_{01} &\stackrel{\text{def}}{=} \mathbb{P}[y_1 = 0, y_2 = 1 | W_{12} = 1] = \lambda e^{-\lambda^+ S} g(2\lambda - \lambda^+) \\
p_{11} &\stackrel{\text{def}}{=} \mathbb{P}[y_1 = y_2 = 1 | W_{12} = 1] = 1 - p_{00} - p_{10} - p_{01}
\end{aligned}$$

where $g(\lambda) = \frac{1-e^{-\lambda S}}{\lambda}$ if $\lambda \neq 0$ and $g(0) = S$.

The probabilities are identified if one observes a sequence of independent draws of such pairs. In this case, the rates are identified since they can be recovered from the probabilities: $\lambda = -\frac{\ln(p_{00})}{2S}$ while λ^+ solves $\frac{\lambda S(p_{00} - e^{-\lambda^+ S})}{\ln(p_{00}) + \lambda^+ S} = p_{10}$, where the left-hand side is strictly decreasing.

Although it is not possible to identify the identity of the first mover – the individual who may have exerted a peer effect on the other – when $y_1 = y_2 = 1$, it is possible to (i) estimate the peer effect strength and (ii) determine the probability of an individual moving first, which provides a probability for each direction of causality.

Example 3. Consider a complete network of size N, *i.e.*, the adjacency matrix corresponds to $\iota\iota' - I$ where ι is a vector of ones. People have baseline rates λ that are updated to λ^{+^k} when k people have opted in.

The outcomes $y_i \in \{0, 1\}$ only inform the fraction of people who opted in, which is insufficient to identify $\{\lambda^{+^k}, k = 0, \ldots, N-1\}$ as only the k-th mover provides a signal about λ^{+^k}. In this regime, positive peer effects cannot be distinguished from stronger baseline rates.[2] Because of the symmetry and perfect connectivity, knowledge about the order of moves is also uninformative.

The complete network case thus requires more information, but can still be identified under additional structure. For instance, if $\lambda = e^\beta$ for some $\beta \in \mathbb{R}$ and

[2] One may deem λ identified through $\mathbb{P}[G = 0]$ insofar as G is observed, though neither this probability nor λ is consistently estimable. Hence, the issue here may also be understood as an identification-estimability gap.

$\lambda^{+^k} = e^{\beta + \frac{k}{N-1}\delta}$ and the times of adoption of early adopters are known, then λ is identified from the behavior of the density around the origin.[3] Given the knowledge of λ, δ is identified from the share of adopters.

The source of non-identification is the lack of repeated signals about rates of a given order when everyone is connected to everyone. This suggests that one can rely on sparser networks to secure identifying information that does not rely on time data or further rate restrictions. This intuition is developed in the next Section. In particular, when the *order* of adoption is known, then the rates can be shown to be identified when degrees are bounded.

The likelihood induced by the stochastic process is available in closed form. This result is the object of the next theorem.

Theorem 2. *If the latent partial times are independent, then the log-likelihood for the exponential specification takes the form*

$$l_N \overset{\text{def}}{=} N^{-1} \ln \left(\sum_{p \in \mathcal{P}} G! \, |\mathcal{P}|^{-1} \left(\prod_{i=1}^{G} (\lambda_{p_i}^{+\{p_1, \cdots, p_{i-1}\}} S) \right) \sum_{g=0}^{G} \frac{e^{-\Lambda_{g|p}}}{\prod_{h \neq g} (\Lambda_{h|p} - \Lambda_{g|p})} \right) \tag{5}$$

where $\mathcal{P}$ contains permutations of the G people who opted in, to each of which the remaining people are appended in any order, and $\Lambda_{g|p} \overset{\text{def}}{=} \sum_{i=g+1}^{N} \lambda_{p_i}^{+\{p_1, \cdots, p_g\}} S$.

The following example shows that the process generalizes i.i.d. exponential draws. The heterogeneity in the rates induces different distributions between individuals, while the peer effects reflected in $\lambda^+ \neq \lambda$ create spatial dependence.

Example 4. Suppose that there is no heterogeneity nor peer effects: $\lambda_i^{+\mathcal{F}} \equiv \lambda$ for all

[3] In this case, identification/estimation relies on the exponential specification because the identifying power of the time of adoptions needs to be used.

i and any $\mathcal{F}$. Then $\Lambda_{g|p} \equiv \lambda(N-g)S$ and the likelihood is

$$
\begin{aligned}
G!\left(\prod_{i=1}^{G}(\lambda_{p_i}^{+\{p_1,\cdots,p_{i-1}\}}S)\right)\sum_{g=0}^{G}\frac{e^{-\Lambda_{pg}S}}{\prod_{h\neq g}(\Lambda_{p_h}-\Lambda_{p_g})} &= G!\,\lambda^G\sum_{g=0}^{G}\frac{e^{-\lambda(N-g)S}}{\prod_{h\neq g}(\lambda(N-h)-\lambda(N-g))} \\
&= G!\,e^{-\lambda(N-G)S}\sum_{g=0}^{G}\frac{e^{-\lambda(G-g)S}}{\prod_{h\neq g}(g-h)} \\
&= e^{-\lambda(N-G)S}\sum_{g=0}^{G}G!\,\frac{e^{-\lambda(G-g)S}}{g!\,(G-g)!}(-1)^{G-g} \\
&= e^{-\lambda(N-G)S}(1-e^{-\lambda S})^G,
\end{aligned}
$$

which is the likelihood when G out of N i.i.d. exponentially distributed variables fall below a cutoff S.

Therefore, probabilities reduce to a standard exponential race with independent draws when the rates do not vary. Changes in exponential rates are a measure of dependence and social interaction; they reflect peer effects.

Remark 3 (2 people example)**.** The probabilities in Example 2 can be directly obtained from the theorem. For instance, noting $\Lambda_{0|p} = (\lambda_1+\lambda_2)S$ and $\Lambda_{1|p} = \lambda_{p_2}^{+}S$, one can verify

$$
\begin{aligned}
\mathbb{P}[y_1=y_2=1|W_{12}=1] = 1 &+ \frac{\lambda_1\lambda_2^{+}e^{-(\lambda_1+\lambda_2)S}}{(\lambda_1+\lambda_2-\lambda_2^{+})(\lambda_1+\lambda_2)} - \frac{\lambda_1 e^{-\lambda_2^{+}S}}{\lambda_1+\lambda_2-\lambda_2^{+}} \\
&+ \frac{\lambda_2\lambda_1^{+}e^{-(\lambda_1+\lambda_2)S}}{(\lambda_1+\lambda_2-\lambda_1^{+})(\lambda_1+\lambda_2)} - \frac{\lambda_2 e^{-\lambda_1^{+}S}}{\lambda_1+\lambda_2-\lambda_1^{+}} \\
&= 1-p_{00}-p_{10}-p_{01}
\end{aligned} \tag{6}
$$

Remark 4 (Computation)**.** Although summing over all permutations can lead to an impractical computational burden, the computational cost can be lowered to practical levels. First, the likelihood factorizes based on the components of W, whose size can be much smaller (classrooms, villages, connected components of friends, etc.). Second, permutations are based on the number of adopters, rather than the size of each component of the network. Hence, the complexity of permutations is substantially reduced, especially when partial information about the ordering is available.

Finally, approximations can further reduce the number of permutations. In par-

ticular, the average over permutations can be estimated by a random sample of permutations by the law of large numbers. The likelihood, score, and Hessian can all be approximated using this technique.

2.4 Identification

This section establishes identification under conditions that guarantee estimability of the rates under two regimes: block structure and a large, sparse network. It is presented in this way to illustrate the key sources of information to allow estimation in our context. Those assumptions are also the building blocks of the forthcoming asymptotics.

The sample comprises outcomes $(y_i, i = 1, \dots, N)$, covariates $(x_i, i = 1, \dots, N)$, and a weighting matrix W. For ease of interpretation, I focus on the common case where W is a row-normalized version of a binary matrix that specifies the peer group of each individual.

Identification relies on our ability to separate information about cross-sectional variation in rates and social influence effects. In the complete network of Example 3, this is not possible without further information because, once a single person opts in, everybody else is subject to social influence; there is no additional information about baseline rates. Most real-life networks, however, are much sparser or exhibit a block structure due to a sampling scheme that targets villages, classrooms, etc. I show how this provides the necessary information under the two regimes: the network consists of components or blocks which do not interact, or are sufficiently sparse. In the latter case, I make use of the assumption that degrees are bounded, which has been invoked in the literature to consider sparser networks (*e.g.*, De Paula, Richards-Shubik and Tamer (2018)), though it can be relaxed to a slow degree growth rate.

Lemma 3. *Let x_i be composed of discrete and continuous variables, and let the rates be continuous in the continuous variables. Assume that latent partial times are independent and that the exponential specification holds with $\lambda_i^{+\mathcal{F}} \in A \subseteq]0, \infty[$, for all i and set $\mathcal{F}$. Then,*
(i) If the network has a block structure with a growing number of blocks $b = 1, \cdots, B$, which are drawn i.i.d. from some population and whose size satisfy $N_b \leq \overline{N}$ for some $\overline{N} \in \mathbb{N}$, then $\lambda_i^{+\mathcal{F}}$ is identified whenever $\mathbb{P}[W_{b;ik} > 0, \forall k \in \mathcal{F}|X_b] > 0$ and $\lambda_j^{+\tilde{\mathcal{F}}} > 0$

for all $j \in \mathcal{F}$, $\tilde{\mathcal{F}} \subseteq \mathcal{F} \backslash \{j\}$.
(ii) If the network becomes large but retains bounded degrees, i.e., $\sup_i d_i \leq \overline{d} < \infty$ *where* $d_i = \sum_{j=1}^{N} \mathbb{1}(W(i,j) > 0)$*, the order in which people opt in is observed, and* A *is compact, then* $\lambda_i^{+\mathcal{F}}$ *is identified for all* i *and any set of people* $\mathcal{F}$ *whose characteristics make a connection with* i *possible with positive probability.*

Remark 5 (Identification of order probabilities)**.** Identification of the family of rates implies that the probability of any sequence of adoption is also identified. Hence, although it is not possible to recover the identity of the first movers or the realized order of adoptions, it is possible to assign a probability to such events.

Remark 6 (Contextual and correlated effects)**.** Contextual peer effects can be straightforwardly allowed for in the large network case. Within a large network, the absence of correlated effects hinges on the (strong) assumption of no unobserved confounders. Thus, under the maintained unconfoundedness condition, information about orderings is sufficient to identify peer effects in full heterogeneity while separating them from correlated effects.

In general, note that no restriction has been placed on the way the specific network heterogeneity may interact with covariates or peer effects. It is thus unsurprising that identification generically fails: separating endogenous effects from contextual or correlated effects requires further information. In the large network case, this can take the form of the order of adoption. With several networks, which induce incidental parameters, this would require functional form restrictions (in the spirit of the SAR reduced form, for instance), distributional assumptions, and/or (partial) information about the order of adoptions.

3 Asymptotic Theory

Although rates are nonparametrically identified, it is crucial to introduce more parsimonious specifications. This avoids severe curse-of-dimensionality issues due to the large number of rates and the dependence on possibly many continuous covariates, and it simplifies interpretation.

A natural model specifies the baseline rates as $\lambda_i = g(x_i, \beta)$ for a positive link function determined by a finite-dimensional vector of parameters β, and lets the

updated rates be obtained by a scaling factor δ. For instance, a simple specification is

$$\ln(\lambda_i^{+\mathcal{F}}) = x_i'\beta + \sum_{j\in\mathcal{F}} W(i,j)\delta. \tag{7}$$

Such a specification reduces the dimensionality of the problem to $(\dim(x_i)+1)$, ensures positive rates, and lets peer effects be described by a parameter δ that reflects how rates are scaled when the members of the peer group opt in. δ controls the strength of spatial dependence.

To the first-order in δ, we have

$$\mathbb{E}[Y_i(\vec{0}) - Y_i(\vec{\infty})|X] \approx \delta S \lambda_i e^{-\lambda_i S} \tag{8}$$

so that δ acts a scaling factor in the change in the probability of adoption induced by the peer group adopting at the onset.

Another way to interpret the peer effect coefficients is through ratios of expected time to adoption. Note that under a log-linear form such as (7), $\lambda_i/\lambda_i^{+\mathcal{F}} = e^{-\delta\sum_{j\in\mathcal{F}} W(i,j)}$ is a ratio of expected waiting times. Using $e^x \approx 1+x$, δ is approximately the proportional change in expected time to adoption going from no to all peer having adopted.

It is easy to relax the parametric specification in various directions. The relevant direction is an applied choice that is likely to differ between applications. For instance, suppose that a researcher is analyzing peer effects in vaccine uptake. They might conjecture that people respond differently depending on the level of education of their peers and that the marginal peer effect is decreasing because the first few movers provide most of the information transmission or reassurance about safety. A natural way to explore heterogeneity or nonlinearity in peer effects is then to add interaction terms such as $\sum_{j\in\mathcal{F}} W(i,j)x_j$, nonlinear terms in $\sum_{j\in\mathcal{F}} W(i,j)$ such as powers, or p-norms in the spirit of Boucher et al. (2024).

For a given parameterization of the rates in terms of a parameter θ (for instance, $\theta = (\beta', \delta)'$ in (7)), estimation proceeds by maximum likelihood. Asymptotics are derived in two frameworks: multiple networks and one large network.

In the first case, I consider a sequence of networks with blocks or components of size N_b, $b = 1, \ldots, B$. With independent blocks, the likelihood factorizes as $l \overset{\text{def}}{=}$

$\ln(\mathbb{P}[Y = y]) = \sum_{b=1}^{B} l_b$, where l_b is the log-likelihood of block b. The logarithmic likelihood of each block is given by the formula established in the theorem of the previous section. The score and Hessian are easily computed in closed form. The details are provided in the Appendix. In the second case, the network consists of a single component whose size grows and a key ingredient comes from knowledge of $\mathcal{F}$, as in the identification condition. In this case, I proceed conditionally on the covariates and the network structure.

In both cases, the estimator $\hat{\theta}$ inherits the usual properties of maximum likelihood estimators: it is consistent and asymptotically normal under regularity conditions. This implies consistency and asymptotic normality of the family of rates. Formally,

Theorem 4 (Consistency)**.** *Let the data-generating process be described by the stochastic process of Section 2.2 and suppose that the family of rates is identified and belongs to the interior of a compact set. Then, (i) if blocks are drawn independently with $W_b \sim \mathcal{D}_W$ such that $N_b < \overline{N}$, the maximum likelihood estimator is consistent with $\hat{\lambda}_i^{+\mathcal{F}} \xrightarrow{p} \lambda_i^{+\mathcal{F}}$, for any $\mathcal{F} \subseteq \{1, \ldots, i-1, i+1, \ldots, N_b\}$, as $B \to \infty$. Alternatively, (ii) if the size of the network grows large and condition (ii) of Lemma 3 is satisfied, the maximum likelihood estimator is consistent with $\hat{\lambda}_i^{+\mathcal{F}} \xrightarrow{p} \lambda_i^{+\mathcal{F}}$, for any $\mathcal{F}$, as $N \to \infty$.*

Moreover,

Theorem 5 (Asymptotic Normality)**.** *Let the data-generating process be described by the stochastic process of Section 2.2 and suppose that the family of rates is identified, belongs to the interior of a compact set, and that the derivatives $\partial_\theta \lambda_i^{+\mathcal{F}}$ are bounded. The Hessians below are well-defined and of full rank.*

Then, if part (i) of the assumptions of the Consistency theorem hold, the maximum likelihood estimator is asymptotically normal as $B \to \infty$ with

$$\sqrt{B}(\hat{\theta} - \theta) \xrightarrow{d} \mathcal{N}\left(0, -H^{-1}\right), \tag{9}$$

where H is the expected block Hessian.

Alternatively, assume that part (ii) of the assumptions of the Consistency theorem hold, then the maximum likelihood estimator is asymptotically normal as $N \to \infty$ with

$$\sqrt{N}(\hat{\theta} - \theta)|X, W \xrightarrow{d} \mathcal{N}\left(0, -H^{-1}\right), \tag{10}$$

provided $\gamma \overset{\text{def}}{=} \lim_{N\longrightarrow\infty} \mathbb{E}[G/N]$ *and the relevant covariances exist. The Hessian is described in the Appendix.*

These theorems allow for standard inference about the underlying parameters θ and transformations of interest by the Delta method. For instance, any rate $\hat{\lambda}_i^{+\mathcal{F}}$ is asymptotically normal with variance $-\partial_{\theta'}\lambda_i^{+\mathcal{F}} H^{-1} \partial_\theta \lambda_i^{+\mathcal{F}}$.

4 Application

Description I revisit the study of Card and Giuliano (2013) on the risky behavior of friends, which makes use of the first two waves of the "The National Longitudinal Study of Adolescent to Adult Health (Add Health)" (Add Health) dataset (Harris et al., 2019). Card and Giuliano (2013) are interested in the estimation of peer effects and their main outcome is the initiation of intercourse among teenagers.

The study provides evidence of peer effects, which is corroborated by a few careful robustness checks, and documents some heterogeneity: peer effects are stronger for female teenagers compared to male teenagers. However, because outcomes are viewed as the result of a simultaneous game, a few concessions had to be made:
- Information at the time of the first survey is difficult to incorporate: people with sexual experience at the time of the first Wave are dropped in the baseline specification.[4]
- Because of the many possible equilibria, they restrict the analysis to best-friend pairs in pure Nash equilibrium and enact equilibrium selection rules.
- Identification relies on a combination of functional form/distributional assumptions and exclusion restrictions.
- Information about the first time of the study's participants, which is part of the questionnaire, is not used to inform the estimates.

I thus review the study through a different lens, incorporating additional information and attempting to shed new light on peer effects in this context. An important feature of the dataset is that partial information on orderings is available: the survey explicitly asked respondents the month/year when they first started having sex. Uncertainty about the ordering is thus limited, but not fully resolved because some

[4] As a robustness check, they propose a control function approach.

people did not report the month and/or year and because first times may have occurred during the same month.[5,6]

Because behavior may exhibit gendered patterns, I estimate gender-specific rates. Friendship links with the opposite gender account for a significant share of friendships—a bit less than 20%—so men and women do not necessarily form two independent networks. As a result, I also consider cross-gender peer effect, *i.e.*, parameters reflecting peer effects on friends of the opposite gender. Specifically, I model

$$\ln(\lambda^{+\mathcal{F}}_{i,\text{male}}) = x_i'\beta_{\text{male}} + \sum_{j\in\mathcal{F}} W_F(i,j)\delta_{\text{male}\leftarrow\text{female}} + \sum_{j\in\mathcal{F}} W_M(i,j)\delta_{\text{male}\leftarrow\text{male}} \tag{11}$$

$$\ln(\lambda^{+\mathcal{F}}_{i,\text{female}}) = x_i'\beta_{\text{female}} + \sum_{j\in\mathcal{F}} W_M(i,j)\delta_{\text{female}\leftarrow\text{male}} + \sum_{j\in\mathcal{F}} W_F(i,j)\delta_{\text{female}\leftarrow\text{female}}, \tag{12}$$

where W_M is the (row-normalized) adjacency matrix of male friends and W_F is the (row-normalized) adjacency matrix of female friends. S is set to 1 year, which corresponds to the time between the two surveys.

The specification thus allows for gender-specific effects of covariates and both own-gender and cross-gender peer effects.

I follow Card and Giuliano (2013) in extracting and constructing relevant covariates from the Add Health questionnaire; see the details therein. The baseline covariates in x_i include age (centered from 10), a race dummy, a measure of physical development, whether the individual lives in a smoking household, a self-reported attitude towards risk and time preference, parental church attendance, and parental education. Dummy variables reflecting missing values of parental education and church attendance are added separately.

Results The main results are presented in Table 4.

[5]Simultaneity is unlikely to be relevant for this application. This can be partially checked from the reported month-year first events: Friends who had their first sexual experience during the same month account for less than 1% of the sample.

[6]The lack of simultaneity also alleviates concerns that friends may coincide with potential sexual partners. Individuals reported friends and romantic partners separately as part of the Add Health questionnaire, which could also be verified to not overlap.

	95% CI (male)	95% CI (female)
Intercept	[-2.67; -0.92]	[-2.65; -1.04]
Age	[0.15; 0.54]	[0.19; 0.57]
Age square	[-0.02; 0.02]	[-0.04; 0]
Black	[0.26; 0.97]	[0.09; 0.66]
GPA	[0.01; 0.03]	[-0.01; 0.01]
Development index	[0.16; 0.38]	[-0.10; 0.12]
Smoking household	[0.20; 0.67]	[-0.02; 0.43]
2-parent household	[-0.80; -0.08]	[-0.75; -0.22]
Parental church attendance	[-0.14; 0.08]	[-0.25; -0.03]
Parents not religious	[-0.23; 0.74]	[-1.35; -0.10]
Church attendance missing	[-0.41; 0.90]	[-1.17; 0.14]
1 parent finished high school	[-0.64; -0.01]	[-0.36; 0.26]
1 parent finished college	[-0.50; -0.01]	[-0.36; 0.11]
Parental education missing	[-0.83; -0.10]	[-0.72; 0.14]
Attitude towards risk	[-0.12; 0.09]	[-0.17; 0.04]
Time preference	[-0.15; 0.04]	[0.01; 0.22]
Peer effect own	[0.03; 0.42]	[0.07; 0.50]
Peer effect cross	[-0.19; 0.36]	[-0.52; 0.33]

Table 1: Estimation results

Similarly to Card and Giuliano (2013), the results indicate that age, race, and physical development (for male) contribute positively to the outcome. There is also no statistical evidence that attitude towards risk and time preference affect the outcome.[7]

There is also evidence of positive peer effects. Point estimates indicate that the average time before the first sexual encounter for male (female) teenagers is reduced by 23% (34%) when all of their male (female) friends have already had sex.

The cross-gender effects are more noisily estimated, as expected from the smaller proportion of friendships that they account for. Although the width of confidence intervals precludes any strong statement in that regard, it appears that individuals respond more strongly to the behavior of a friend when they share the same gender– there is no strong evidence that cross-gender peer effects matter.

Additional results and discussion I explore alternative specifications to assess the robustness of the results and to gather additional insight into the structure of peer effects. First, I consider quadratic peer effect terms to account for a potential

[7]Note that attitude towards risk is a self-reported score as of Wave I. As such, it may respond to recent behavior prior to the survey, which may include decisions about the outcome of interest.

decreasing influence in peer effect strength.

The coefficients associated with the squared averages are too noisily estimated to draw any definitive conclusion, though the negative point estimates and the higher first-order coefficients may suggest that the first peers to initiate intercourse exert the strongest influence.

Within large components, correlated effects can be allowed for, *e.g.*, by estimating group-specific coefficients or including group-specific intercepts.

However, estimating group-specific coefficients may be unreliable with some of the components in which fewer people experienced their first time during the sampling period. Furthermore, this would not account for unobserved confounders that vary within components. To attempt to alleviate such concerns, I augment the baseline specification of the individual rates by multiplying each lambda by e^{ε_i} where the ε_i are allowed to be correlated when people are linked.

Overall, the results corroborate the findings of Card and Giuliano (2013) without relying on functional form restrictions for identification. Peer effects are important drivers of teen behavior and the effect may be slightly more pronounced for women.

Although the shock-augmented specification may alleviate some concerns about omitted confounders or correlated shocks, some threats to the validity of the estimates remain, especially in the form of the usual caveats about survey data. Another concern is that people do not necessarily share details about their sexual experience with all of their friends. Typically, this would lead to an attenuation bias: the coefficients capture a mix of actual peer effects and an absence of reaction due to imperfect information. If there are differential rates of sharing for different gender combinations, this may influence the relative magnitude of the various δ's. It is unlikely to be a solvable issue from a modeling perspective, however. Collecting additional data on the type of information that is shared with different friends could be a more promising avenue.

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Appendix A: Proofs and Further results

Probabilities in Example 2

Proof. The result is shown for possibly different rates λ_1, λ_2. Consider the probability $\mathbb{P}[y_1 = 1, y_2 = 0]$. This corresponds to

$$\begin{aligned}
\mathbb{P}[T_1 \leq S, T_2 > S] &= \int_0^S p[T_1 = t, T_2 > S] \, dt \\
&= \int_0^S \lambda_1 e^{-\lambda_1 t} \, p[T_2 > S | T_1 = t] \, dt \\
&= \int_0^S \lambda_1 e^{-\lambda_1 t} \, p[T_2^1 > t | T_1 = t] \, p[T_2^2 > S - t | T_1 = t] \, dt \qquad (13) \\
&= \int_0^S \lambda_1 e^{-\lambda_1 t} \, p[T_2^1 > t] \, p[T_2^2 > S - t | T_1 = t] \, dt \\
&= \int_0^S \lambda_1 e^{-\lambda_1 t} e^{-\lambda_2 t} e^{-\lambda_2^+ (S-t)} \, dt
\end{aligned}$$

where densities are denoted by p.

If $\lambda_1 + \lambda_2 - \lambda_2^+ = 0$, this is simply $\lambda_1 S e^{-\lambda_2^+ S}$. If $\lambda_1 + \lambda_2 - \lambda_2^+ \neq 0$, then integrating yields $\lambda_1 e^{-\lambda_2^+ S}(1 - e^{-(\lambda_1 + \lambda_2 - \lambda_2^+)S})/(\lambda_1 + \lambda_2 - \lambda_2^+)$.

Then, $\mathbb{P}[y_1 = 0, y_2 = 1]$ follows by symmetry, $\mathbb{P}[y_1 = 0, y_2 = 0] = e^{-(\lambda_1 + \lambda_2)S}$ is trivial, and $\mathbb{P}[y_1 = 1, y_2 = 1]$ is deduced by the fact that the four probabilities sum up to 1.

□

Proof of Theorem 2

Proof. The log-likelihood is given by

$$\ln(\mathbb{P}[Y_1 = y_1, \ldots, Y_N = y_N | X, W]) \qquad (14)$$

Assume without loss that the individuals who opted in correspond to the first G

observations. Then, summing over different orderings,

$$\begin{aligned}&\mathbb{P}[y_1 = \ldots = y_G = 1; y_{G+1} = \ldots = y_N = 0]\\&= \sum_{p\in\mathcal{P}} \mathbb{P}[y_1 = \ldots = y_G = 1; y_{G+1} = \ldots = y_N = 0; T_{G+1}, \cdots, T_N > S; T_{p_1} < \ldots < T_{p_G}]\end{aligned}$$

where the sum is over all permutations (with generic element $p = (p_1, \ldots, p_G)$) of the G first arrivals.

Consider the representative term in the sum when individual i is the i-th to opt in. Using the independence of partial times and the exponential specification and following steps similar to the 2-individual examples, this term is given by

$$\begin{aligned}&\int_0^S \int_{t_1^1}^\infty \cdots \int_{t_1^1}^\infty \prod_{i=1}^N \lambda_i e^{-\lambda_i t_i^1} \quad \int_0^{S-t_1^1} \cdots \int_{t_2^2}^\infty \prod_{i=2}^N \lambda_1^{+1} e^{-\lambda_i^{+1} t_i^2} \cdots\\&\int_0^{S-\sum_{g=1}^{G-1} t_g^g} \cdots \int_{t_G^G}^\infty \prod_{i=G}^N \lambda_i^{+1,\cdots,G-1} e^{-\lambda_i^{+1,\cdots,G-1} t_i^G}\\&\int_{S-\sum_{g=1}^G t_g^g}^\infty \cdots \int_{S-\sum_{g=1}^G t_g^g}^\infty \prod_{i=G+1}^N \lambda_i^{+1,\cdots,G} e^{-\lambda_i^{+1\cdots+G} t_i^{G+1}}\\&dt_1^1 \cdots dt_N^1 dt_2^2 \cdots dt_N^2 \cdots dt_G^G \cdots dt_N^G \cdots dt_1^{G+1} \cdots dt_N^{G+1}\end{aligned}$$

which, after some algebra, reduces to

$$e^{-\sum_{i=G+1}^N \lambda_i^{+1,\cdots,G} S} \left(\prod_{i=1}^G \lambda_i^{+1,\cdots,i-1} S \right) I_{\{c_i, i=1,\cdots,G\}} \tag{15}$$

with $I_{\{h_i, i=1,\cdots,H\}} \stackrel{\text{def}}{=} \frac{1}{\prod_{g=1}^G \dot{\Lambda}_g} + (-1)^G \sum_{g=1}^G \frac{1}{\prod_{h\neq g}(\dot{\Lambda}_g - \dot{\Lambda}_h)} \frac{e^{-\dot{\Lambda}_g}}{\dot{\Lambda}_g}$ and $\dot{\Lambda}_g \stackrel{\text{def}}{=} \sum_{i=g}^N \lambda_i^{+1,\cdots,g-1} S - \sum_{i=G+1}^N \lambda_i^{+1,\cdots,G} S$.

Introducing $\dot{\Lambda}_{G+1} = 0$ and then $\Lambda_g \equiv \Lambda_{g|12\cdots G} = \dot{\Lambda}_{g+1} + \sum_{i=G+1}^N \lambda_i^{+1,\cdots,G} S = \sum_{i=g+1}^N \lambda_i^{+1,\cdots,g} S \geq 0$ (with equality only if $g = G = N$), the probability simplifies to[8]

[8] As in the example with 2 individuals, it can be checked from the integral computation that the limit for a term in a denominator converging to 0 gives the correct probability at the points where the function above is undefined. These cases correspond to knife-edge cases in which a move does not change the rate at which the next event occurs: enhanced rates due to peer effects exactly offset the withdrawal from the current mover.

$$\prod_{i=1}^{G}(\lambda_i^{+1,\cdots,i-1}S)\sum_{g=0}^{G}\frac{e^{-\Lambda_g}}{\prod_{h\neq g}(\Lambda_h-\Lambda_g)} \tag{16}$$

□

Proof of Lemma 3

Proof. The proof considers the two cases separately.

Identification under block structure In what follows, $\vec{0}$ denotes a 0-vector of conformable size. Consider the identified probability $\mathbb{P}[Y=\vec{0}|X=x,W] = e^{-\sum_{i=1}^{n}\lambda_i(x_i)S}$. Setting $x_i = x\ \forall i$, $\lambda(x) = \frac{-\ln(p_{00}(x,\ldots,x))}{nS}$ and thus the baseline rates are identified.

Then, $p_{10}(x_1,x_2) = \lambda_1 e^{-\lambda_2^+ S} g(\lambda_1+\lambda_2-\lambda_2^+)$ allows us to identify λ_2^+ nonparametrically as $\lambda^+(x_2,x_1)$ because $e^{-xS}g(\lambda_1+\lambda_2-x) = e^{-(\lambda_1+\lambda_2)S}g(-(\lambda_1+\lambda_2-x))$ is an invertible function of x for any λ_1,λ_2.

Consider now

$$\begin{aligned}\mathbb{P}[y_1=1,y_{-1}=0|X,W] =& \lambda_1\left(\frac{e^{-\sum_{i=2}^{N}\lambda_i^{+1}S}}{\sum_{i=1}^{N}\lambda_i-\sum_{i=2}^{N}\lambda_i^{+1}}-\frac{e^{-\sum_{i=1}^{N}\lambda_i S}}{\sum_{i=1}^{N}\lambda_i-\sum_{i=2}^{N}\lambda_i^{+1}}\right)\\ =& \lambda_1 e^{-\sum_{i=1}^{N}\lambda_i S} g\left(\sum_{i=2}^{N}\lambda_i^{+1}-\sum_{i=1}^{N}\lambda_i\right)\end{aligned}$$

where g is defined as in Example 2.

Setting $x_j = x$ for all $j\neq 1$, $\mathbb{P}[y_1=1,y_{-1}=0|X=(x_1,x,\ldots,x),W] = \lambda_1 e^{-\sum_{i=1}^{N}\lambda_i S} g\left(\sum_{j:W_{j1}\neq 0}(\lambda_j^{+1}-\lambda_j)-\lambda_1\right)$. By letting $W_{k1}=0$, $k\neq j$ and letting W_{j1} vary, $\lambda_j^{+1} = \lambda_j^{+1}(W_{j1})$ is identified by invertibility of $g(\cdot)$ and identification of the baseline λ_j.

Proceeding iteratively, $\lambda_i^{+\mathcal{F}}(W)$ is identified from the conditional probabilities and the knowledge of all $\lambda_i^{+\tilde{\mathcal{F}}}(W)$ with $|\tilde{\mathcal{F}}|<|\mathcal{F}|$.

Identification under sparsity The proof again proceeds iteratively by showing how to identify baseline rates, then (once-) updated rates under the knowledge of baseline rates, etc. Covariates are handled by conditioning throughout and adjusting values to match the target rate (taking them as discrete, and using increasingly finer

grids and continuity of rates to handle the continuous case), as in the proof under block structure. Note that W takes on finitely many values since it corresponds to the row-normalization of a binary matrix with bounded degrees.[9]

Note that we observe at least K adopters for any $K \in \mathbb{N}$ with probability approaching one. Since there are finitely many values of W, there is a finite number of $\lambda^{+\mathcal{F}}$ so that each rate can be mapped into the set $\{1, \ldots, n_\lambda\}$ by some function m.

Let the index of the rate of the k-th adopter (at their adoption time) be denoted by Γ_k. For any $K \in \mathbb{N}$, λ is given by the inverse of $f_K(\lambda) = \frac{1}{K}\sum_{k=1}^{K} \mathbb{P}(\Gamma_k = m(\lambda))$, which is strictly increasing in λ if the network is connected. This function is identified from $\frac{1}{K}\sum_{k=1}^{K} \mathbb{1}(\Gamma_k = m(\lambda)) - \mathbb{P}(\Gamma_k = m(\lambda)) \to^p 0$ by a law of large numbers, noting the conditional independence and integrability.

□

Proofs of Theorem 4 and 5

I divide the proof into two sections depending on the type of asymptotics: several networks and one large network.

Asymptotics with several networks

Proof. The proofs verify Newey and McFadden (1994)'s sufficient conditions for consistency and asymptotic normality of extremum estimators.

The parameter space is compact by assumption. The log-likelihood of block b is $l_b = \sum_{y \in \{0,1\}^{N_b}} \mathbb{1}[Y_b = y] \ln(\mathbb{P}[Y_b = y])$, where the elements of the sum are bounded from below using that $N_b \leq \overline{N}$ and compactness of the rates and from above by 0. Since in addition $\mathbb{P}[Y_b = y]$ is continuous, a uniform law of large numbers implies (i) $\sup_{\{\lambda\}} |B^{-1}\sum_b l_b - \mathbb{E}[l_b]| \to 0$ and (ii) $\mathbb{E}[l_b]$ is continuous.

By continuity of the Hessian matrix and compactness of the parameter space, it follows that

$$\sup_{\{\lambda\}} |H_n - H| \to 0 \tag{17}$$

where H is continuous. Asymptotic normality of the score follows from a central limit theorem for triangular arrays, noting that $\mathbb{V}[l_b] \to \sigma^2$.

[9] A similar construction may be used to show identification for arbitrary weighting matrices under continuity of the rates in the corresponding entries of W.

By assumption, the parameters are identified and live in a compact set. The objective function converges uniformly to $\mathbb{E}[l_b]$, which is continuous, by uniform law of large numbers (*e.g.*, Lemma 2.4 in Newey and McFadden (1994)). Indeed, l_b is continuous and

$$\min_{y,p,\lambda} \mathbb{P}[p(\lambda)] \leq e^{l_b} \leq 1, \tag{18}$$

where the minimization is over a finite set for y, p since $N_b \leq \overline{N}$ and the minimization over λ is over a compact set for a continuous function, so that $\sup|B^{-1}\sum_{b=1}^{B} l_b - \mathbb{E}[l_b]| \to 0$.

$$\frac{1}{\sqrt{B}} \sum_{b=1}^{B} s_b \to^d \mathcal{N}(0; \mathbb{E}[s_b s_b']) \tag{19}$$

noting that the score has mean zero and

$$\begin{aligned}
\mathbb{V}[s_b] &= \mathbb{E}[s_b s_b'] \\
&\leq C\mathbb{E}[\partial L_b \partial L_b'] \\
&= C \sum_{W_b} \mathbb{P}[W_b] \mathbb{E}[\partial L_b \partial L_b' | W_b] \\
&= C \sum_{W_b} \mathbb{P}[W_b] \sum_{\tilde{y}} \mathbb{P}[Y = \tilde{y}] \partial L_b(\tilde{y}) \partial L_b'(\tilde{y}) \\
&\leq C \max_{\tilde{y}} \partial L_b(\tilde{y}) \partial L_b'(\tilde{y})
\end{aligned} \tag{20}$$

□

Proof. For an ordering, say, $1, \ldots, G$, we have

$$\prod_{i=1}^{G} \lambda_i^{+1,\cdots,i-1} S \sum_{g=0}^{G} \frac{e^{-\Lambda_g}}{\prod_{h\neq g} \Lambda_h - \Lambda_g} = \sum_{g=0}^{G} \frac{e^{-R_g}}{\prod_{h\neq g} R_h - R_g} \tag{21}$$

where $R_g \stackrel{\text{def}}{=} \Lambda_g - \sum_{i=1}^{G} \ln(\lambda_i^{+1,\cdots,i-1} S)$

The sum can be recognized as the G-th divided difference of the exponential function e^{-x}, multiplied by $(-1)^G$. By the Hermite-Genocchi formula, noting that the G-th derivative of e^{-x} is $(-1)^G e^{-x}$, we obtain

$$\sum_{g=0}^{G} \frac{e^{-R_g}}{\prod_{h\neq g} R_h - R_g} = \int_{\Delta_G} \exp\left(-\sum_{g=0}^{G} t_g R_g\right) \mathrm{d}t \tag{22}$$

where Δ_G is the G-dimensional simplex.

Define the integral on the scaled simplex as a function F:

$$F(u) \stackrel{\text{def}}{=} \int_{t_g \geq 0, \sum_{g=0}^{G} t_g = u} \exp\left(-\sum_{g=0}^{G} t_g R_g\right) \mathrm{d}t \tag{23}$$

The Laplace transform of F has a closed form:

$$\mathcal{L}F(s) = \prod_{g=0}^{G} \frac{1}{R_g + s} \tag{24}$$

The probability of interest, which coincides with $F(1)$, can thus be obtained from an inverse Laplace transform, and the convergence can be interpreted as a saddle

point problem:

$$\begin{aligned}\int_{\Delta_G} \exp\left(-\sum_{g=0}^{G} t_g R_g\right) \mathrm{d}t &= F(1) \\ &= \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \frac{e^s}{\prod_{g=0}^{G}(R_g+s)}\mathrm{d}s \\ &= \frac{1}{2\pi i N^G}\int_{c_0-i\infty}^{c_0+i\infty} \frac{(e^z)^N}{\prod_{g=0}^{G}(R_g/N+z)}\mathrm{d}z \\ &= \frac{1}{2\pi i N^G}\int_{c_0-i\infty}^{c_0+i\infty} \left(e^{z-\frac{1}{N}\sum_{g=0}^{G}\ln(R_g/N+z)}\right)^N \mathrm{d}z\end{aligned} \tag{25}$$

Deforming the contour of integration to go through the real saddle point and applying the Laplace principle, it is seen that the integral concentrates around the z that satisfies the stationarity condition $\frac{1}{N}\sum_{g=0}^{G}\left(R_g/N\right)+z)^{-1}=1$.

This implies that, for the observed p, the log-likelihood is given by

$$\begin{aligned}l_N = z_N^*(p) &- \frac{1}{N}\sum_{g=0}^{G}\ln\left(R_{g|p}/N + z_N^*\right) \\ &+ \frac{G}{N}\ln\left(\frac{G}{N}\right) - \frac{G}{N} + O(N^{-1}\ln(N)) + O(G^{-1}\ln(G))\end{aligned} \tag{26}$$

where z_N^* solves $\frac{1}{N}\sum_{g=0}^{G}\left(R_{g|p}/N+z_N^*\right)^{-1}=1$.

Using the envelope theorem, the score is given by

$$s_N \overset{\text{def}}{=} -\frac{1}{N}\sum_{g=0}^{G}\frac{\partial_\theta R_{g|p}/N}{R_{g|p}/N+z_N^*(p)} + O(N^{-1}\ln(N)) \tag{27}$$

while the Hessian is

$$\begin{aligned}H_N \overset{\text{def}}{=} -\frac{1}{N}\sum_{g=0}^{G}&\left(\frac{\partial_{\theta\theta'}R_{g|p}/N}{R_{g|p}/N+z_N^*(p)} - \frac{\partial_\theta R_{g|p}/N\partial_{\theta'}R_{g|p}/N}{(R_{g|p}/N+z_N^*(p))^2} - \frac{\partial_\theta R_{g|p}/N\partial_{\theta'}z_N^*(p)}{(R_{g|p}/N+z_N^*(p))^2}\right) \\ &+ O(N^{-1}\ln(N)),\end{aligned} \tag{28}$$

where $\partial_{\theta'}z_N^*(p) = -\left(\frac{1}{N}\sum_{g=0}^{G}\frac{1}{(R_{g|p}/N+z_N^*(p))^2}\right)^{-1}\frac{1}{N}\sum_{g=0}^{G}\frac{\partial_{\theta'}R_{g|p}/N}{(R_{g|p}/N+z_N^*(p))^2}$ by applying

the implicit function theorem on the condition $\frac{1}{N}\sum_{g=0}^{G}\left(R_{g|p}/N + z_N^*(p)\right)^{-1} = 1$.

The proof proceeds by a classical expansion around the true value of θ, we have

$$\begin{aligned}\sqrt{N}(\hat{\theta}-\theta) &= -\left(H_N\right)^{-1}\sqrt{N}s_N \\ &\xrightarrow{d} \mathcal{N}(0,-H^{-1}),\end{aligned} \tag{29}$$

since, as argued below, $\sup_\theta |H_n - H| \xrightarrow{p} 0$, the score converges in distribution to a normal random variable, and the information equality applies.. In what follows, the reference to the permutation is dropped again to ease notation.

The variable R_g, and similarly the derivative $\partial_\theta R_g$, can be decomposed as

$$R_g = \sum_{h=1}^{g}\Delta_h + R_0 = \sum_{h=1}^{g}(\Delta_h - \mathbb{E}[\Delta_h|\mathcal{F}_{h-1}]) + \sum_{h=1}^{g}(\mathbb{E}[\Delta_h|\mathcal{F}_{h-1}] - \mathbb{E}[\Delta_h]) + \mathbb{E}[R_g] + R_0 - \mathbb{E}[R_0] \tag{30}$$

with $\Delta_h \overset{\text{def}}{=} R_h - R_{h-1}$. The terms in each sum are bounded and uncorrelated and the last term corresponds to $(R_0 - \mathbb{E}[R_0]) = -\sum_{i=1}^{G}\ln(\lambda_i^{+1,\ldots,i-1}) + \mathbb{E}\left[\ln\left(\lambda_i^{+1,\ldots,i-1}\right)\right] = O_p(N^{1/2})$. The law of large numbers then implies convergence of R_g/N to its expectation. Furthermore, the convergence is uniform in θ since the terms are continuous in θ and uniformly bounded by $2\sup_{i,\mathcal{F}}\left|-\lambda_i^{\mathcal{F}} + \sum_j \lambda_j^{+\mathcal{F}\cup\{i\}} - \lambda_j^{+\mathcal{F}}\right| \leq ((\overline{d}+1)\overline{\lambda} - \overline{d}\underline{\lambda})$, where $\overline{\lambda} = \sup_{i,\mathcal{F}}\lambda_i^{+\mathcal{F}}$ and $\underline{\lambda} = \inf_{i,\mathcal{F}}\lambda_i^{+\mathcal{F}}$.

As a result, by letting

$$\hat{r}(g/N) \overset{\text{def}}{=} \begin{pmatrix} R_g \\ \partial_\theta R_g \end{pmatrix}/N, \tag{31}$$

where the functions are extended in the usual way to the unit interval by interpolating, and defining $r(z) = \lim_{N\to\infty}\mathbb{E}[\hat{r}(z)]$, we have

$$\sup_\theta |\hat{r}(z) - r(z)| \xrightarrow{p} 0. \tag{32}$$

In addition,

$$\sup_\theta |z_N^* - z^*| \xrightarrow{p} 0 \tag{33}$$

because z_N^* is the minimizer of a strictly convex function which converges uniformly to the function that z^* minimizes.

Note that, up to discretization error,

$$\int_0^{G/N} \frac{1}{\hat{r}(z) + z_N^*} \mathrm{d}z = 1. \tag{34}$$

Expanding around r, z^*, γ gives

$$\sqrt{N}(z_N^* - z^*) = \frac{\frac{\sqrt{N}(G/N-\gamma)}{r_1(\gamma)+z^*} - \int_0^\gamma \frac{\sqrt{N}(\hat{r}_1(z)-r_1(z))}{(r_1(z)+z^*)^2}\,\mathrm{d}z}{\int_0^\gamma \frac{1}{(r_1(z)+z^*)^2}\,\mathrm{d}z} + o_p(1). \tag{35}$$

Now, the score satisfies

$$\sqrt{N}\frac{1}{N}\sum_{g=0}^{G} \frac{\partial \hat{r}}{\hat{r} + z_N^*} = \sqrt{N}\int_0^{G/N} \frac{\partial \hat{r}}{\hat{r} + z_N^*} + o_p(1). \tag{36}$$

At the true parameter, the limiting score is such that $\int_0^\gamma \frac{\partial_\theta r(z)}{r(z)+z^*}\,\mathrm{d}z = 0$. Then, applying a first-order expansion to the numerator, denominator, and integration endpoint in $\int_0^{G/N} \frac{\partial \hat{r}}{\hat{r}+z_N^*}$ yields

$$\begin{aligned}
\sqrt{N}s_N = &- \int_0^\gamma \frac{\sqrt{N}(\hat{r}_{2:(n_\theta+1))}(z) - r_{2:(n_\theta+1))}(z))}{r_1(z) + z^*}\,\mathrm{d}z \\
&+ \int_0^\gamma \frac{r_{2:(n_\theta+1))}(z))\sqrt{N}(\hat{r}_1(z) - r_1(z))}{(r_1(z) + z^*)^2}\,\mathrm{d}z \\
&+ \sqrt{N}(z_N^* - z^*)\int_0^\gamma \frac{r_{2:(n_\theta+1))}(z))}{(r_1(z) + z^*)^2}\,\mathrm{d}z \\
&- \sqrt{N}(G/N - \gamma)\frac{r_{2:(n_\theta+1))}(\gamma))}{r_1(\gamma) + z^*} + o_p(1).
\end{aligned} \tag{37}$$

Then, using $\sqrt{N}\begin{pmatrix} \hat{r} - r \\ G/N - \gamma \end{pmatrix} \xrightarrow{d} \rho^*$, where ρ^* is a centered, continuous Gaussian process as shown below, the continuous mapping theorem implies

$$\sqrt{N}s_N \xrightarrow{d} \mathcal{N}(0, \Omega) \tag{38}$$

where Ω is the limiting variance matrix, induced by the structure in (37). In view of the information equality, $\Omega = -H$.

Finally, using the uniform convergence results on $\hat{r}$ and z_N^*, we have

$$\sup_\theta |H_N - H| \xrightarrow{p} 0 \tag{39}$$

where

$$H \stackrel{\text{def}}{=} -\int_0^\gamma \frac{\partial_{\theta\theta'} r(z)}{r(z)+z^*}\mathrm{d}z + \int_0^\gamma \frac{\partial_\theta r(z)\partial_{\theta'} r(z)}{(r(z)+z^*)^2}\mathrm{d}z - \frac{\int_0^\gamma \frac{\partial_\theta r(z)}{(r(z)+z^*)^2}\mathrm{d}z \int_0^\gamma \frac{\partial_{\theta'} r(z)}{(r(z)+z^*)^2}\mathrm{d}z}{\int_0^\gamma \frac{1}{(r(z)+z^*)^2}\mathrm{d}z}. \tag{40}$$

It remains to establish convergence to the process ρ^*. The main steps are sketched below.

Time processes The first key step is to establish joint convergence to processes in the time domain, say, on the interval $[0, \overline{S}]$ with $\overline{S} \stackrel{\text{def}}{=} S + 1$. Specifically, consider

$$\sqrt{N}\begin{pmatrix} A_N - a \\ B_N - b \\ \partial_\theta B_N - b_\theta \\ C_N - c \\ \partial_\theta C_N - c_\theta \end{pmatrix} \xrightarrow{d} \begin{pmatrix} \mathcal{A} \\ \mathcal{B} \\ \mathcal{B}_\theta \\ \mathcal{C} \\ \mathcal{C}_\theta \end{pmatrix}, \tag{41}$$

where $A_N(t) \stackrel{\text{def}}{=} \frac{1}{N}\sum_i \mathbb{1}\{T_i \le t\}$, $B_N(t) \stackrel{\text{def}}{=} \frac{S}{N}\sum_{i:T_i>t} \lambda_i^{+\mathcal{F}_t}$, and $C_N(t) \stackrel{\text{def}}{=} \frac{1}{N}\sum_{i:T_i\le t} \ln(S\lambda_i^{+\mathcal{F}_{T_i^-}})$, $a, b, b_\theta, c, c_\theta$ are the limits of the corresponding expectations, and $\mathcal{A}, \mathcal{B}, \mathcal{B}_\theta, \mathcal{C}$, and $\mathcal{C}_\theta$ are centered processes, Gaussian processes on $D([0, \overline{S}])$ whose covariance kernels are given through
$K_A(s,t) \stackrel{\text{def}}{=} \lim_{N\to\infty} \frac{1}{n}\sum_{i=1}^N \sum_{j=1}^N \mathbb{C}(a_i(s), a_j(t) \mid X, W)$,
$K_B(s,t) \stackrel{\text{def}}{=} \lim_{N\to\infty} \frac{1}{n}\sum_{i=1}^N \sum_{j=1}^N \mathbb{C}(b_i(s), b_j(t) \mid X, W)$,
$K_{B_\theta}(s,t) \stackrel{\text{def}}{=} \lim_{N\to\infty} \frac{1}{n}\sum_{i=1}^N \sum_{j=1}^N \mathbb{C}(b_{\theta,i}(s), b_{\theta,j}(t) \mid X, W)$,
$K_C(s,t) \stackrel{\text{def}}{=} \lim_{N\to\infty} \frac{1}{n}\sum_{i=1}^N \sum_{j=1}^N \mathbb{C}(c_i(s), c_j(t) \mid X, W)$, and
$K_{C_\theta}(s,t) \stackrel{\text{def}}{=} \lim_{N\to\infty} \frac{1}{n}\sum_{i=1}^N \sum_{j=1}^N \mathbb{C}(c_{\theta,i}(s), c_{\theta,j}(t) \mid X, W)$, where
$a_i(t) \stackrel{\text{def}}{=} \mathbb{1}(\{T_i \le t\})$, $b_i(t) \stackrel{\text{def}}{=} S\mathbb{1}(T_i > t)\lambda_i^{+\mathcal{F}_t}$, $b_{\theta,i}(t) \stackrel{\text{def}}{=} S\mathbb{1}(T_i > t)\partial_\theta \lambda_i^{+\mathcal{F}_t}$, $c_i(t) \stackrel{\text{def}}{=} \mathbb{1}(T_i \le t)\ln(S\lambda_i^{+\mathcal{F}_{T_i^-}})$, $c_{\theta,i}(t) \stackrel{\text{def}}{=} \mathbb{1}(T_i \le t)\partial_\theta \ln(S\lambda_i^{+\mathcal{F}_{T_i^-}})$.

Convergence of finite-dimensional distributions follows from a dependency-graph CLT (Janson, 1988) applied to suitably truncated objects, as shown below. For ease of exposition, consider $B_N(t) = \frac{1}{n}\sum_{i=1}^N b_i(t)$ with $b_i(t) \stackrel{\text{def}}{=} S\sum_{i:T_i>t} \lambda_i^{+\mathcal{F}_t}$. The general

case follows by (i) stacking the elements from A_n, $\partial_\theta B_N$, C_N, and $\partial_\theta C_N$ and (ii) applying the same argument to arbitrary linear combinations of $b_i(t_1), \ldots, b_i(t_k)$ to establish joint Gaussian convergence at finitely many times.

One can construct a truncated version of the process by shutting down the influence of m-th and higher-order neighbors around each i. The influence of those neighbors is not more than $O(\sum_{k \geq m} \frac{(\overline{d}\overline{\lambda}t)^k}{k!})$ by bounding the probability of a sequence of m reactions. Letting $m = m_N$ evolve at a suitable rate, noting the maximal degree satisfies $D_N = O((\overline{d}+1)^{2m_N})$ can be made to grow arbitrarily slowly, one can apply the dependency graph CLT (Janson, 1988) (with variables pertaining to i and j independent whenever their distance exceeds $2m$) to the truncated version of $b_i(t)$ to show asymptotic normality: $\sqrt{N}(B_N(t) - \mathbb{E}[B_N(t) \mid X, W]) \xrightarrow{d} \mathcal{N}(0, K_B(t,t))$.

Regarding tightness, we have $\mathbb{P}[b_i(t) \neq b_i(s) | X, w] \leq (1+\overline{d})\overline{\lambda}|t-s|$ since only decisions of i and their neighbors can affect b_i. Since the influence of m-th order neighbors vanishes faster than exponentially in m, one can bound

$$\begin{aligned}
&\mathbb{P}[|\sqrt{N}(B_N(t) - \mathbb{E}[B_N(t)|X,W]) - \sqrt{N}(B_N(s) - \mathbb{E}[B_N(s)|X,W]))|\varepsilon|X, w] \\
&\leq \mathbb{E}[(\sqrt{N}(B_N(t) - \mathbb{E}[B_N(t)|X,W]) - \sqrt{N}(B_N(s) - \mathbb{E}[B_N(s)|X,W]))^4|X,w]\varepsilon^{-4} \\
&= O(|t-s|/N)
\end{aligned} \tag{42}$$

Inversion The overall rate at time t is given by $\sum_{i:T_i>t} \lambda_i^{+\mathcal{F}_t} = \frac{N}{S} B_N(t)$. As a result, $M_N(t) = N A_N(t) - \frac{N}{S}\int_0^t B_N(u)\,\mathrm{d}u$ is a counting-process martingale, with predictable quadratic variation $\langle M_N \rangle_t = \frac{N}{S}\int_0^t B_N(u)\,\mathrm{d}u$.

In particular, we have $\partial_t \mathbb{E}[A_N(t) \mid X, W] = \frac{\mathbb{E}[B_N(t)|X,W]}{S}$. This ensures invertibility: $\partial_t \mathbb{E}[A_N(t) \mid X, W] = \frac{1}{n}\sum_{i=1}^N \mathbb{E}[\mathbb{1}\{T_i > t\}\lambda_i^{+\mathcal{F}_t} \mid X, W] \geq \underline{\lambda} e^{-\overline{\lambda}\overline{S}} > 0$ for $t \leq \overline{S}$.

Then with $\mathbb{E}[A_N(t) \mid X, W] \to a$ and $\mathbb{E}[B_N(t) \mid X, W] \to b$ sufficiently regularly, we obtain $a'(t) = b(t)/S > 0$. The calendar-time adoption fraction therefore has a strictly increasing deterministic limit on every compact time interval. We can then consider the generalized inverse

$$\tau_N(z) = \inf\{t : A_N(t) \geq z\} \tag{43}$$

whose limit is $\tau(z) = a^{-1}(z)$.

From a quantile-process delta method (Van Der Vaart and Wellner, 1996),

$$\sqrt{N}(\tau_N(z)-\tau(z)) = -\sqrt{N}\frac{A_N(\tau(z))-a(\tau(z))}{a'(\tau(z))}+o_p(1) = -\sqrt{N}S\frac{A_N(\tau(z))-a(\tau(z))}{b(\tau(z))} \tag{44}$$

Limiting process for $\hat{r}$ and G/N Expanding to account for the time variation, we have

$$B_N(\tau_N(z))-b(\tau(z)) = B_N(\tau_N(z))-b(\tau_N(z))+b(\tau_N(z))-b(\tau(z)). \tag{45}$$

Taylor expansion of the second term and the inverse formula yield

$$\sqrt{N}\left(B_N(\tau_N(z))-b(\tau(z))\right) \xrightarrow{d} \mathcal{B}(\tau(z))-\frac{b'(\tau(z))}{a'(\tau(z))}\mathcal{A}(\tau(z)), \tag{46}$$

i.e., the fluctuation at a given adoption index consists of the fluctuation of B_N at a fixed calendar time, minus the correction caused by the random time required to reach that index.

Noting that $R_g/N = B_N(t_g) - C_N(S)$, we have

$$\sqrt{N}(\hat{r}_1(z)-r_1(z)) \xrightarrow{d} \mathcal{B}(\tau(z))-\frac{b'(\tau(z))}{b(\tau(z))}\mathcal{A}(\tau(z))-\mathcal{C}(S). \tag{47}$$

Similarly,

$$\sqrt{N}(\hat{r}_{2:(n_\theta+1)}(z)-r_{2:(n_\theta+1)}(z)) \xrightarrow{d} \mathcal{B}_\theta(\tau(z))-\frac{b'_\theta(\tau(z))}{b(\tau(z))}\mathcal{A}(\tau(z))-\mathcal{C}_\theta(S), \tag{48}$$

and finally

$$\sqrt{N}(G/n-\gamma) = \sqrt{N}(A_N(S)-a(S)) \xrightarrow{d} \mathcal{A}(S), \tag{49}$$

where the convergence results also hold jointly by linearity.

□